

Module 2 — Quiz A (Def & Notation)

Question bank · After Read 1 · open notes · unlimited attempts

Module 2, Quiz A: Definitions & notation

Reading: [Module 2 Read 1](#)

Project: [Project 2](#) · Canvas: Def & Notation quiz (Module 2)

Work these practice items after [Read 1](#). Quiz A covers test vs CI, proportion notation (n , N , p , $\hat{p}$), binomial μ/σ , Wald CI by hand (multiplier 2), conditions, and standardized z . Section G is optional Type I/II enrichment.

! By-hand multiplier 2 (Quiz A & B)

On these quizzes, use 2 as the critical value for Wald proportion CIs and approximate mean CIs unless a item says otherwise. Your endpoints may be slightly different from `prop.test()` ($z^* \approx 1.96$) or `t.test()` (t^* depends on n). That is expected. [Synthesis](#), labs, and Project 2 use R output.

i Conditions (three layers)

Before any formula: (1) SRS from a numbered frame; proportions also need BRP (BINS). (2) Large-sample / model: proportion $\rightarrow n\hat{p}$ and $n(1 - \hat{p})$ both ≥ 10 (else exact/Plus Four); mean $\rightarrow$ population roughly normal or $n > 30$. (3) Population size: finite N with non-tiny $n/N \rightarrow$ FPC; very large / “infinite” population $\rightarrow$ no FPC. See the [conditions checklist](#) in Read 2.

Sampling, means, FPC, R output $\rightarrow$ [Quiz B](#) then [Synthesis](#) after the [exercise lab](#).

Section A — Confidence intervals & test vs CI (Read 1 §2)

A1. 95% confidence level

Define 95% confidence level in terms of repeated sampling and intervals.

A2. Confidence interval

Define confidence interval in one or two clear sentences.

A3. Test vs CI — research questions

A fair-coin study observes 7 heads in 16 flips.

- (a) What question does a hypothesis test of $H_0 : p = 0.5$ answer?
- (b) What question does a 95% CI for p answer?

A4. Test vs CI — wording

Which pair best matches test vs confidence interval?

- A) *What is* the population proportion? / *Is* the proportion equal to 0.30?
- B) *Is* the proportion equal to 0.30? / *What is* a plausible range for the population proportion?

C) Both ask the same question with different notation.

A5. Incorrect CI interpretation

Which is wrong for a single 95% CI you already computed?

A) If we repeated sampling many times, about 95% of such intervals would contain the parameter.

B) There is a 95% probability the parameter lies in this interval.

C) I am 95% confident the true population proportion is between the endpoints.

A6. Test vs CI standard error

One-proportion test uses p_0 in the SE; one-proportion CI uses $\hat{p}$ in the SE. Why?

Section B — Notation: n , N , p , x , $\hat{p}$ (Read 1)

B1. Part-time job survey

Random sample $n = 120$ undergraduates; $x = 46$ work ≥ 10 hrs/week.

Identify parameter, statistic, and compute $\hat{p}$.

B2. Clinic satisfaction

$n = 80$ patients sampled; $x = 52$ very satisfied.

Compute $\hat{p}$ and state what p represents.

B3. n vs N

A campus has $N = 4,200$ students; you measure $n = 80$ in an SRS.

What does each letter denote?

B4. p vs $\hat{p}$

In a vaccine study, 88 of 200 patients showed a response.

Identify x , n , $\hat{p}$, and the parameter p .

B5. Parameter

Define parameter in one or two clear sentences (proportion or mean example).

B6. Statistic

Define statistic in one or two clear sentences (proportion or mean example).

Section C — Binomial μ , σ & normal approximation (Read 1 §3)

C1. Binomial mean

ESP test: $n = 25$ trials, $p = 0.2$. What are μ and σ for $X = \text{number correct}$?

C2. Normal approximation conditions

A binomial count has $n = 100$, $p = 0.04$. Can you use the normal approximation with $np \geq 10$ and $n(1 - p) \geq 10$? Explain briefly.

C3. Empirical Rule

State the Empirical Rule for a normal distribution (three parts: 68%, 95%, 99%).

C4. Rejection region sketch

Under a normal null curve for a one-proportion test, where is the rejection region for a two-sided test at roughly $\alpha = 0.05$ using the Empirical Rule? How does shaded tail area relate to p-value?

C5. Binomial SD

ESP test: $n = 25$, $p = 0.2$. Compute $\sigma = \sqrt{np(1 - p)}$ for $X =$ number correct.

C6. Exact vs normal — small counts

$n = 16$, test $H_0 : p = 0.5$; under H_0 , $np = 8$ and $n(1 - p) = 8$. Should you rely on the normal approximation?

Section D — Wald 95% CI for p by hand (Read 1 §3)

D1. Wald proportion conditions

For Wald 95% CI $\hat{p} \pm 2\sqrt{\hat{p}(1 - \hat{p})/n}$, name sampling, BINS, and large-count requirements.

D2. St. Joe's Wald CI

St. Joe's: $n = 200$ patients, $x = 14$ deaths. Compute $\hat{p}$, $SE \approx 0.018$, and approximate 95% Wald CI using multiplier 2.

D3. Wald CI — compute endpoints

SRS $n = 220$, $x = 80$ with a part-time job. $\hat{p} = 80/220 \approx 0.364$, $SE \approx 0.032$. Compute approximate 95% CI using $2 \times SE$.

D4. Exact when Wald fails

$n = 18$, $x = 2$ successes; $n\hat{p} < 10$. Which CI approach is safer than Wald?

D5. Plus Four trigger

St. Joe's small sample: $n = 16$, $x = 3$ deaths. Why might you use Plus Four instead of Wald?

Section E — Standardized z for one proportion (Read 1 §4)

E1. Coin test hypotheses

16 heads in 20 flips; test if coin is fair (two-sided). State H_0 , H_a , and define p in words.

E2. Role of p_0

In $z = (\hat{p} - p_0)/\sqrt{p_0(1 - p_0)/n}$, what is p_0 ?

E3. Compute z

16 heads in 20 flips; $H_0 : p = 0.5$. Compute $z = (\hat{p} - 0.5) / \sqrt{0.5(0.5)/20}$ (decimal). Is $|z|$ beyond 2?

E4. Descriptive vs inferential (proportion)

One binary variable from a sample: give one descriptive summary and one inferential tool for estimation.

Section F — Inferential statistics & sampling distribution**F1. Inferential statistics**

Define inferential statistics in one or two sentences and contrast with descriptive statistics.

F2. Sampling distribution

Define the sampling distribution of $\hat{p}$ in one sentence.

Section G — Type I/II vocabulary**G1. Significance level**

Define the significance level α in hypothesis testing.

G2. Type I error

Define a Type I error in hypothesis testing.

G3. Type II error

Define a Type II error in hypothesis testing.
